

TSB_20260303

Date: 3 March 2026

Applies to: All Siemens 3VA molded case circuit breakers installed in EPEC/QT switchboards and related assemblies.

Subject: Acceptable conductor terminations for Siemens molded case circuit breakers.

Purpose

This bulletin clarifies acceptable conductor termination methods for Siemens 3VA molded case circuit breakers (MCCBs) equipped with standard UL-listed mechanical lugs. It is being issued to prevent overheating, connection failure, and equipment damage resulting from the use of unapproved terminations.

Siemens 3VA Lug Design and Listing

Siemens 3VA mechanical lugs supplied for UL applications are designed and tested for:

- Direct compression of appropriately sized copper or aluminum conductors
- Conductor classes specified in Siemens documentation and lug markings
- Installation using manufacturer-specified torque values

Mechanical lugs in MCCBs are evaluated under UL 489 as part of the breaker assembly. Any deviation from listed conductor types or termination methods can place the connection outside its tested configuration.

Acceptable Termination Methods

Unless explicitly approved by Siemens for the specific lug kit and conductor class, the following practices should be observed:

Acceptable:

- Direct insertion of properly sized stranded conductor within mechanical lug rating
- Use of listed compression lugs or terminals where specified
- Installation per Siemens torque and preparation requirements

Figure 1 below is a Siemens table detailing approved compression lug connection types. Figures 2a-2c details approved conductors for mechanical lugs. For clarity, IEC values that do not apply to applications North America have been struck out, and UL appropriate values are highlighted.

Unacceptable Termination Methods

Solid pin terminals are unlisted and **NOT APPROVED** for these breakers, as the mechanical lugs are designed for direct conductor compression. The use of solid pin terminals can create the following conditions:

- The screw cannot properly deform and seat into the conductor bundle
- Contact area may be reduced
- Clamping force can diminish under thermal cycling
- Localized heating and connection failure may result

Use of solid pin terminals will void any equipment warranties, express or implied. **If you have any questions about the suitability of a desired connection, please reach out to support@epecsolutionsinc.com**



Connection / Anschluss / Raccordement / Conexión / 连接

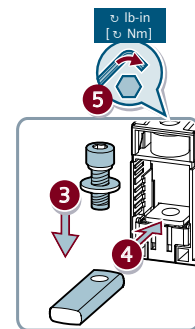
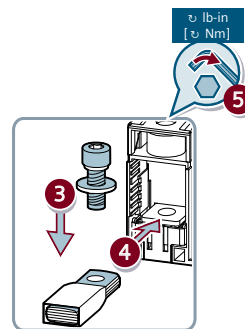
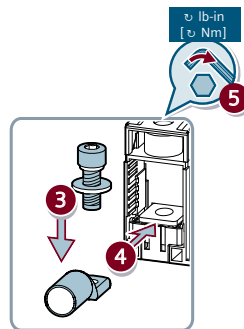
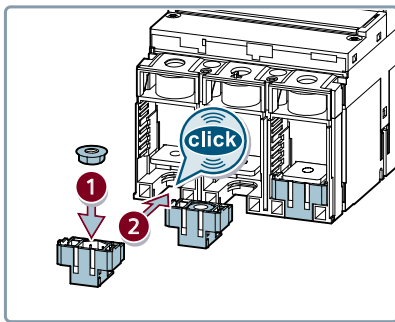


						
3VA4/5	3VA41 / 51	A			B	C
3VA5	3VA52	A			B	C
	3VA53 / 54	A			B	C / D
3VA6	3VA61 / 62	A			B	C
	3VA63 / 64	A			B	C / D
3VA914.-0K.00	150 / 250	A			B	C

A

B

C



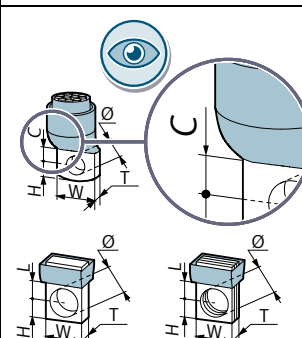
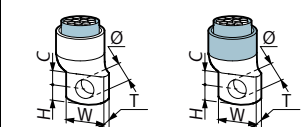
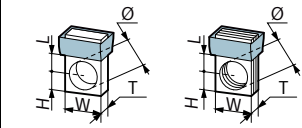
		$W_{\max}$ in [mm]	T in [mm]	L in [mm]	$C_{\min}$ in [mm]	$\varnothing$ in [mm]	H in [mm]		
3VA41		0.66 [17]	≤ 0.25 [≤ 6.5]	0.32 - 0.39 [8 - 10]	0.32 [8]	1/4 [6.5]	0.31 [8]	5/32"	70 [8]
3VA51		0.66 [17]	≤ 0.25 [≤ 6.5]	0.32 - 0.39 [8 - 10]	0.32 [8]	1/4 [6.5]	0.31 [8]	5/32"	70 [8]
3VA52		1 [25]	≤ 0.31 [≤ 8]	0.40 - 0.47 [10 - 12]	0.40 [10]	5/16 [8.5]	0.39 [10]	1/4"	177 [20]
3VA61 / 62 3VA53 / 54 3VA63 / 64		1.37 [35]	0.04 - 0.39 [1 - 10]	0.67 - 0.78 [17 - 20]	0.59 [15]	3/8 [11]	0.57 [14.5]	5/16"	354 [40]
3VA914.-0K.00 150 / 250		1 [25]	≤ 0.31 [≤ 8]	0.40 [10]		5/16 [8.5]	0.39 [10]	1/4"	177 [20]

Figure 1

Connection / Anschluss / Raccordement / Conexión / 连接							
		 in [mm]	 UL [IEC]		 AWG/kcmil [mm²]	 in [mm]	 lb-in [N·m]
 TA1.0	3VA41 3VA51	5/32" [4]	Cu	 {Class 5/6}	[1.5 - 10]	0.5 [12.2]	14 AWG: 20 12 - 10 AWG: 35 8 AWG: 55 [1.5 - 2.5 mm²: 2.3] [4 - 6 mm²: 4] [10 mm²: 6.2]
				 {Class 5/6}	[1.5 - 10]		
				 Class B/C [Class 2]	14 - 8 [1.5 - 10]		
				 Class A [Class 1]	14 - 8 [1.5 - 10]		
			Al	 Class B	8		
				 Class A	12 - 10		
 TC1.0	3VA41 3VA51	5/32" [4]	Cu	 Class B/C	14 - 8	0.5	14 AWG: 20 12 - 10 AWG: 35 8 AWG: 55
				 Class A	14 - 10		
 TA1.1	3VA41 3VA51	5/32" [4]	Cu	 {Class 5/6}	[10 - 35]	0.5 [12.2]	 3VA913.-0JB1. *E01* 3VA913.-0JG1. *E01* 8 AWG: 55 3 AWG - 3/0: 100 [10 mm²: 6.2] [16 mm² - 95 mm²: 11.3]  3VA913.-0JB1. *E02>* 3VA913.-0JG1. *E02>* 8 AWG - 6 AWG: 55 4 AWG - 1/0: 70 2/0 - 3/0: 100 [10 mm² - 16 mm²: 6.2] [25 mm² - 50 mm²: 7.9] [70 mm² - 95 mm²: 11.3]
				 {Class 5/6}	[10 - 70]		
				 Class B/C [Class 2]	8 - 3/0 [10 - 95]		
				 {Class 1}	[10 - 16]		
			Al	 Class B [Class 2]	8 - 3/0 [10 - 95]		
				 {Class 1}	[10 - 16]		
 TC1.1	3VA41 3VA51	5/32" [4]	Cu	 Class B/C	14 - 2/0	0.5	14 AWG: 25 12 AWG: 35 10 AWG - 2/0: 55
				 Class A	14 - 10		
 TA1.3	3VA52 3VA61 3VA62	5/32" [4]	Cu	 {Class 5/6}	[1.5 - 35]	0.7 [18]	14 AWG - 8 AWG: 55 6 AWG - 2 AWG: 75 1 AWG - 1/0: 120 [1.5 - 10 mm²: 6.2] [16 - 35 mm²: 8.5] [50 mm²: 13.6]
				 {Class 5/6}	[1.5 - 35]		
				 Class B/C [Class 2]	14 - 1/0 [1.5 - 50]		
				 Class A [Class 1]	14 - 10 [1.5 - 16]		
			Al	 Class B	8 - 1/0		
 TC1.3	3VA52 3VA61 3VA62	5/32" [4]	Cu	 Class B/C	14 - 1/0	0.5	14 AWG - 8 AWG: 55 6 AWG - 2 AWG: 75 1 AWG - 1/0: 120

Figure 2a



Connection / Anschluss / Raccordement / Conexión / 连接



		 In [mm]		UL [IEC]	 AWG/kcmil [mm²]	 in [mm]	 ft-lb-in [N·m]
 TA1.4	3VA52 3VA61 3VA62	5/16" [8]	Cu	 {Class 5/6}	{25 – 150}	0.8 [20]	6 AWG - 2 AWG: 140 1 AWG - 350 kcmil: 275 [16 - 35 mm²: 15.8] [50 - 185 mm²: 31.1]
				 {Class 5/6}	{16 – 120}		
				 Class B/C {Class 2}	6 - 350 {16 – 185}		
				 {Class 1}	{16}		
			Al **	 Class B {Class 2}	6 - 350 {16 – 185}		
				 {Class 1}	{16}		
 TC1.4	3VA52 3VA61 3VA62	5/16" [8]	Cu	 Class B/C	6 - 350	0.8	6 AWG - 2 AWG: 140
 TA1.5	3VA53 3VA54	5/16" [8]	Cu	 {Class 5/6}	{50 – 240}	1.1 [28]	375 [42.4]
	3VA63 3VA64			 {Class 5/6}	{50 – 240}		
	Cu wire 380A Al wire 310A			 Class B/C {Class 2}	1 - 600 {50 – 300}		
			Al	 Class B {Class 2}	1 - 600 {50 – 300}		
 TC1.5	3VA53 3VA54 3VA63 3VA64	5/16" [8]	Cu	 Class B/C	2 - 600	1.1	500

**)	EN	When connecting Al-cable: use antioxidant.	DE	Bei Anschluss von Al-Kabel: Antioxidationsmittel verwenden.
	FR	En cas de raccordement d'Al-câble: utiliser un antioxydant.	ES	Cuando conecte el cable de aluminio: utilice antioxidante.
	IT	Quando si collega il cavo Al: usare un antiossidante.	PT	Ao ligar o cabo Al: Usar antioxidante.
	TR	Aluminyum kabloyu bağladiğında, antioksidan (pas önleyici) kulan.	PY	При подключении Al-кабеля: используйте антиоксидант.
	PL	Przy podłączaniu kabla Al: należy użyć środka przeciwutleniającego.	中	连接铝制电缆时：使用抗氧化剂

Figure 2b



Connection / Anschluss / Raccordement / Conexión / 连接



		 in [mm]		UL [IEC]	 AWG/kcmil [mm ²]	 in [mm]	 lb-in [Nm]	
 TA4.1	3VA55 3VA65 3VA66	5/16" [8]	Cu	 {Class 5/6}	{240}	L1, L2: 1.12 [28.5]	400 - 750 kcmil: [240 - 300 mm ²]	375 42.4
				 {Class 5/6}	{240}			
				 Class B/C [Class 2]	400 - 750 [240 - 300]			
				 Class K	500			
			Al	 Class B	500 - 750			
 TA4.2	3VA55 3VA65 3VA66	5/16" [8]	Cu	 {Class 5/6}	{240}	L1: 1.13 [28.7] L2: 1.69 [42.9] L3: 2.30 [58.4]	500 - 750 kcmil: [240 - 300 mm ²]	375 42.4
				 {Class 5/6}	{240}			
				 Class B/C [Class 2]	500 - 750 [240 - 300]			
				 Class K	500			
			Al	 Class B [Class 2]	500 - 750 [240 - 300]			
 TA4.3	3VA55 3VA65 3VA66	5/16" [8]	Cu	 {Class 5/6}	{185}	L1, L2: 0.91 [23.1] L3, L4: 1.75 [44.5]	4/0 - 500 kcmil: [120 - 240 mm ²]	325 36.7
				 {Class 5/6}	{120 - 150}			
				 Class B/C [Class 2]	4/0 - 500 [120 - 240]			
			Al	 Class B [Class 2]	4/0 - 500 [120 - 240]			
 TC4.3	3VA55 3VA65 3VA66	5/16" [8]	Cu	 {Class 5/6}	{185}	L1, L2: 0.91 [23.1] L3, L4: 1.75 [44.5]	4/0 - 500 kcmil: [120 - 240 mm ²]	325 36.7
				 {Class 5/6}	{120 - 150}			
				 Class B/C [Class 2]	4/0 - 500 [120 - 240]			
 TA4.4	3VA55 3VA65 3VA66	5/16" [8]	Cu	 {Class 5/6}	{150 - 185}	L1, L2: 1.13 [28.7] L3, L4: 2.2 [56.1]	4/0 - 600 kcmil: [120 - 300 mm ²]	375 42.4
				 {Class 5/6}	{150 - 185}			
				 Class B/C [Class 2]	4/0 - 600 [120 - 300]			
			Al					
				 Class B [Class 2]	4/0 - 600 [120 - 300]			

Figure 2c